Peek, don't fetch.

A JOIN returns every matching pair. EXISTS only asks whether a match is there, so a row can never repeat.

ONE ROW PER CLUB · THREE QUERIES

"Which clubs have members?" and "which have none?"

CLUBS where we start	MEMBERS students.club_id = club_id	INNER JOIN fetches every pair	WHERE EXISTS asks: any member?	WHERE NOT EXISTS asks: no member?
Photography	Aarav Priya	Photography Photography dup	Photography ✓ 2 members, still 1 row	⊗ dropped
Robotics	Kabir	Robotics	Robotics ✓	⊗ dropped
Drama	Vikram	Drama	Drama ✓	⊗ dropped
Debate	*no members*	⊗ dropped	⊗ dropped	Debate ✓
OUTPUT SIZE		4 rows. Photography twice, so you need DISTINCT.	3 rows. One per club, no DISTINCT needed.	1 row. Only the empty club survives.

THE TWO PATTERNS

Semi-join: active clubs

```
SELECT c.club_name
FROM clubs c
WHERE EXISTS (
  SELECT 1 FROM students s
  WHERE s.club_id = c.club_id
);
```

Anti-join: empty clubs

```
SELECT c.club_name
FROM clubs c
WHERE NOT EXISTS (
  SELECT 1 FROM students s
  WHERE s.club_id = c.club_id
);
```

EXISTS = keep me if a match is found. **NOT EXISTS** = keep me if none is. Start from `students` instead to find the club-less: **Sneha**.

WHICH ONE DO I NEED?

Columns from **both** tables	JOIN
Only **yes/no**: does a match exist?	EXISTS
Find the **orphans**: no match at all	NOT EXISTS